

CU-423 RoboLink

Welcome to the Wireless World

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What are we doing today?

1

Understand

What is
communication
engineering

2

Connect

Bluetooth and BLE

3

Configure

HC-05 BLE and AT
mode

4

Compete

today's robot race

Big idea: Understanding how to communicate with Arduino wirelessly.

What Is Communication Engineering?

Communication engineers design:

- devices that send and receive messages
- signals that travel through air, wires, or light
- ways to reduce mistakes from noise and interference
- rules so devices understand each other



Examples

Text message

Wi-Fi video call

TV remote

RoboLink phone app

Same job every time: move a message clearly enough that the destination can use it.

A Basic Communication System



The channel can change the signal.

distance, walls, interference, and noise

Engineers measure capacity, delay, and errors.

Our Robot Communication System



Command map

1 = Stop 2 = Forward 3 = Backward
4 = Left 5 = Right

Lab connection

Module TXD -> Arduino D10
Arduino D11 -> module RXD through a voltage divider

What Is Bluetooth?

- uses radio waves instead of wires
- works over short distances, usually across a room
- uses the 2.4 GHz radio band
- sends data in small packets using agreed rules
- connects phones, headphones, keyboards, sensors, and robots



Bluetooth®

**Bluetooth is not the internet.
It is the short wireless link between nearby
devices.**



Why Is It Called Bluetooth?

Harald "Bluetooth" Gormsson

- King of Denmark and Norway more than 1,000 years ago
- remembered for uniting groups of people
- engineers wanted their new radio link to unite different devices
- the name started as a code name and stayed

The name comes from a real historical king, not from a blue phone icon.



The logo combines old runic letters for H and B.

H = Harald
B = Bluetooth

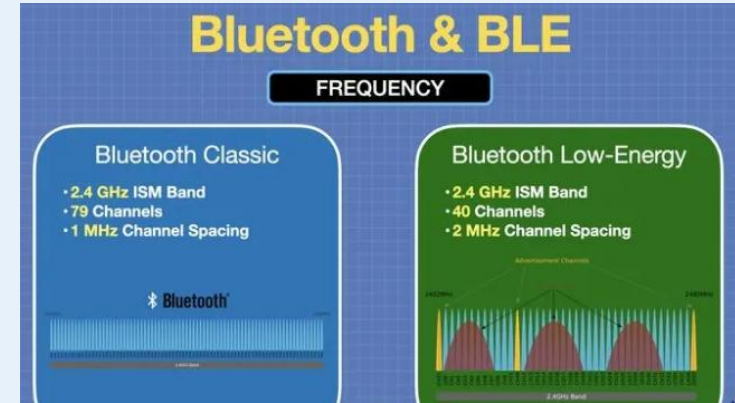
Two Bluetooth Protocols: SPP and BLE

SPP: Serial Port Profile

- acts like a wireless serial cable
- common in older Bluetooth Classic robot projects
- sends a stream of bytes after pairing

BLE: Bluetooth Low Energy

- designed for low-power devices and sensors
- uses services and characteristics
- good for short robot control messages



Our classroom module and app use BLE.

SPP: A Wireless Serial Cable

Easy mental model

USB serial cable without the cable

- Arduino sends and receives characters
- phone app opens a serial connection
- good for simple terminal apps

Important for today

- Standard HC-05 modules are usually Bluetooth Classic SPP modules.
- Our classroom board is an HC-05-style BLE edition.
- When online tutorials disagree with the lab sheet, follow the lab sheet.

SPP = Serial Port Profile

BLE: Bluetooth Low Energy

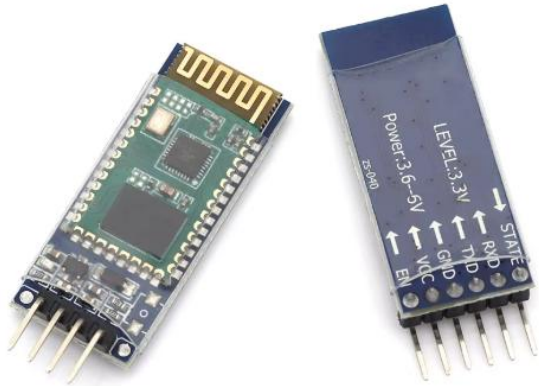
- data is organized into services and characteristics
- great for sensors, beacons, fitness bands, and robot commands
- uses less energy than keeping a large data stream open all the time

Our BLE command idea

- 1 Stop
- 2 Forward
- 3 Backward
- 4 Left
- 5 Right

The app sends one byte.
The Arduino decides what the byte means.

Meet the HC-05 BLE Module



HC-05-style BLE edition

What it does

- receives a Bluetooth/BLE radio signal from the phone
- turns wireless data into UART serial data
- sends bytes to Arduino through TXD and RXD pins
- lets the robot be controlled without USB

Normal mode = data mode

Setup = AT mode

HC-05 Specs to Know

Radio

2.4 GHz Bluetooth radio
with built-in antenna

Serial

UART TXD/RXD pins
Lab baud: 9600

Connection caution

Protect module RXD
with a voltage divider

Modes

Normal data mode
and AT command mode

Common classic HC-05 datasheets list Bluetooth v2.0+EDR, SPP, 3 Mbps modulation, typical -80 dBm sensitivity, and up to +4 dBm transmit power.

Master and Slave

Classic Bluetooth wording

- Master starts or controls the connection.
- Slave waits to be found and connected.
- Many HC-05 AT commands use ROLE=1 for master and ROLE=0 for slave.

Master -> Slave

Newer BLE wording

- Central usually scans and connects. The phone is often central.
- Peripheral advertises and waits. The robot module is often peripheral.
- Same idea, more modern terms.

Phone central -> module peripheral

What Is AT Mode?

In normal data mode:

- phone commands go straight to the Arduino
- the robot can drive
- the module acts like a wireless cable

In AT mode:

- the module listens for text commands that begin with AT
- we can read or change module settings

Useful AT questions

AT

Are you alive? -> OK

AT+VERSION?

Firmware version

AT+NAME?

Module name

AT+ADDR?

Bluetooth address

AT+ROLE?

Master/slave role

AT+UART?

Serial baud settings

How to Access AT Mode

- 1 Disconnect the phone/app from the module.
- 2 Power off the module or Arduino.
- 3 Connect EN high as your board requires.
- 4 Power on while KEY/EN is active. A slow LED blink usually means command mode.
- 5 Upload the AT_test code.
- 6 Open Serial Monitor at the AT baud rate, often 38400 or 9600, with Both NL & CR.
- 7 Send AT. If the module replies OK, you are in AT mode.

Module Pin	Arduino Nano Pin	Notes
TXD	D10	Module transmits to Arduino receive pin.
RXD	D11	Use a voltage divider before RXD.
GND	GND	Must share ground with the Arduino.
VCC	5V or 3.3V	Use the correct power for your module board.
EN	5V	Connect EN to 5V.

To DO Task

- Change the module name to **Group #**
- Get the **module Address** and write it down on the sticker
- **Stick** the sticker to your module
-

HC-05 Bluetooth Module AT Commands

Task	Command / Action	Expected Reply	Notes
Enter AT mode	Connect EN/KEY to 5V, then power the module	LED blinks slowly	Put the HC-05 into AT mode before sending commands.
Test AT mode	AT	OK	Confirms the module is ready.
Check current name	AT+NAME?	+NAME:<name> then OK	Shows the current Bluetooth module name.
Change Bluetooth name	AT+NAME=RoboLink	OK	Replace RoboLink with the name you want.
Get module address	AT+ADDR?	+ADDR:xxxx:xx:xxxxxx then OK	Returns the Bluetooth address.
Restart module	AT+RESET	OK	Useful after changing the name.

Typical AT mode settings: 38400 baud, line ending set to Both NL & CR.

Lab Path: Test First, Then Drive

Part 1: Wireless command test

- phone sends one command
- module passes it to Arduino D10/D11
- Arduino prints it in Serial Monitor
- built-in LED turns on/off as a quick proof

Goal: prove wireless messages arrive correctly

Part 2: RoboLink control

- same commands now control the L9110 motor driver
- TouchDown sends movement command
- TouchUp sends 1 so the robot stops
- motor speed can be tuned if the robot drifts

Goal: control motion safely and reliably

Break!



Robot Race Today

Race mission

- connect the phone to your team robot
- drive through the course without touching the robot
- use controlled moves, not full-speed guessing
- fastest clean run advances in the cup

Race rules

- one driver at a time
- hands off the robot during the run
- reset only from the start line
- if Bluetooth disconnects, stop and reconnect before retrying
- respect other teams' space and wiring

Engineering target: reliable control beats random speed.

7-Group Race Cup



Stage	Match	Team A	Team B	Winner
Round 1	A	Group 1	Group 2	
Round 1	B	Group 3	Group 4	
Round 1	C	Group 5	Group 6	
Bye	-	Group 7	Advances	Group 7
Semifinal	SF1	Winner A	Winner B	
Semifinal	SF2	Winner C	Group 7	
Final	F	Winner SF1	Winner SF2	
Champion	-	Winner F		

Exit Ticket

1

In our robot system, what is the source and what is the destination?

2

What part of the system is the channel?

3

How is BLE different from SPP?

4

What does AT mode let us check or change?

One-sentence summary: Wireless control works when every part of the communication chain works.

The end-of-course survey link is now open! Please take a few minutes to complete the feedback form.



Break!



Good luck in the RoboLink race.

